CarbonLedger

Scope 1, 2 and 3 Emissions Report

Meridian Electronics Manufacturing Co., Ltd.

Reporting period: 2026-01-01 to 2026-06-30

GWP version: AR6

Track: Sustainability & Social Impact

Factor year distribution (factors used): 2025: 1, 2026: 19

Total emissions: 610.455 t CO2e

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1. Cover

Reporting entity: Meridian Electronics Manufacturing Co., Ltd.
Reporting period: 2026-01-01 to 2026-06-30
Generated on: 2026-10-14
GWP version: AR6 (100-year global warming potentials; see chapter 3)
Track: Sustainability & Social Impact
Factor year distribution: 2025: 1 factors used, 2026: 19 factors used
Disclaimer: See chapter 10

2. Executive summary

Total emissions for the reporting period amount to 610.455 t CO2e.

Scope	t CO2e	Share
Scope 1	160.186	26240.4%
Scope 2	208.784	34201.4%
Scope 3	241.485	39558.2%

Top emission source	t CO2e
District heat supply for office campus (2,880 GJ converted)	208.784
Employee commuting rail season tickets, aggregate H1 2026	188.46
Refrigerant HFC-32 top-up for plant HVAC	148.8

1. Prioritize the largest emission sources first: District heat supply for off commuting rail season tickets, aggregate H1 2026; Refrigerant HFC-32 the majority of the reported footprint, so procurement and logistics actio reductions per unit of effort.
2. Adopt supplier-specific data collection for the top suppliers identified factors progressively; each replaced factor lowers the uncertainty level o
3. Institutionalize quarterly re-runs of this accounting pipeline so that ne ledger, keeping the report current and allowing reduction progress to be

3. Accounting method and boundary

This report follows the GHG Protocol Corporate Value Chain (Scope 3) Accounting and Reporting Standard in conjunction with the Corporate Standard. Organizational boundary: operational control. Operational boundary: Scope 1 direct, Scope 2 purchased energy, and Scope 3 value-chain categories as identified per record.

GWP version: AR6, sourced from IPCC assessment report 100-year values. Cross-GWP-version factor usage: 20 of 20 records (100.0%) (each flagged in the interface and downgraded to at least medium uncertainty).

Factor data year	Factors used
2025	1
2026	19

Data quality overall assessment: Data quality is moderate: 20 of 20 reco
version different from the task setting and are flagged accordingly.

4. Emissions breakdown

Scope	t CO2e
Scope 1	160.186
Scope 2	208.784
Scope 3	241.485

Scope 3 category	t CO2e
Cat-1 Purchased goods and services	30.468
Cat-2 Capital goods	5.862
Cat-3 Fuel- and energy-related activities	9.252
Cat-4 Upstream transportation and distribution	4.915
Cat-5 Waste generated in operations	0.02
Cat-6 Business travel	2.509
Cat-7 Employee commuting	188.46
Cat-8 Upstream leased assets	0
Cat-9 Downstream transportation and distribution	0
Cat-10 Processing of sold products	0
Cat-11 Use of sold products	0
Cat-12 End-of-life treatment of sold products	0
Cat-13 Downstream leased assets	0
Cat-14 Franchises	0
Cat-15 Investments	0

Supplier (top 10)	t CO2e
Sinopec	11.386
Shenzhen CoolTech	148.8
China Southern Power Grid	0
Nanshan Heat Group	208.784
Aluworks Supply	27.924
Unspecified	200.832
Wuhan PCB Fab	0.003
Hanwei Machine Tools	5.862
COSCO	4.868
Shenzhen Environmental Services	0.019

5. Activity data detail

Source	Description	Quantity	Factor	t CO2e	Uncertainty
doc_cb4cf26e#page=1	Diesel fuel (average biofuel blend) for own fleet, combustion	14000 kg	EF-D26-WTTFUELS-015	11.386	high
doc_293da9de#page=1	Refrigerant HFC-32 top-up for plant HVAC	12 kg	EF-D26-REFRIGERANTOTHER-004	148.8	medium
doc_4c587592#page=1	Factory grid electricity, H1 2026	1250000 kWh	EF-D26-BUSINESSTRAVELAIR-018	0	high
doc_a1463419#page=1	District heat supply for office campus (2,880 GJ converted)	799997.76 kWh	EF-E25-FUEL-045	208.784	high
doc_5c95444c#page=1	Aluminium sheet 5052 for chassis panels	9500 kg	EF-D26-FUELS-007	27.924	high
doc_37858e03#page=1	PET film, food grade	1200 kg	EF-D26-WASTEDISPOSAL-113	0.006	high
doc_37858e03#page=1	PP injection pellets	2600 kg	EF-D26-WASTEDISPOSAL-010	0.003	high
doc_37858e03#page=1	LDPE packaging film	900 kg	EF-D26-WASTEDISPOSAL-111		

					0.008	high
doc_f805bf37#page=1	4-layer PCB assemblies (12 lots)	640 kg	EF-D26-WASTEDISPOSAL-006		0.003	high
doc_6d3c4595#page=1	CNC machining centre VM-850 (capital equipment)	8200 kg	EF-D26-WTTFUELS-031		5.862	high
doc_21c4bca0#page=1	Well-to-tank upstream emissions of fleet diesel (matches D-01 load)	14000 kg	EF-D26-WTTFUELS-017		9.252	high
doc_429d057d#page=1	FCL ocean freight of sensor modules, Yantian to Hamburg	17300 kg	EF-D26-SEAFREIGHT-034		4.868	high
doc_2e0c5faa#page=1	Inbound road freight, Guangzhou to Nanshan (ambient rigid truck)	6800 kg	EF-D26-WTTROADFREIGHT-017		0.046	high
doc_b8dcb9c1#page=1	Landfill disposal of mixed production waste	4200 kg	EF-D26-WASTEDISPOSAL-080		0.02	high
doc_ced2c2aa#page=1	Return economy flight SZX-HAJ for trade fair, 1 passenger	17600 km	EF-D26-BUSINESSTRAVELAIR-019		2.509	high
doc_1bbf2968#page=1	Heat-treated wooden export pallets (ISPM-15)					

		520 kg	EF-D26-MATERIALUSE-070	0.545	high	
doc_42f74b9b#page=1						
	Employee commuting rail season tickets, aggregate H1 2026	210000 km	EF-D26-DELIVERYVEHICLES-012	188.46	medium	
doc_b03c3c53#page=1						
	International express courier, Shenzhen to Austin	85 kg	EF-D26-WASTEDISPOSAL-134	0.001	high	
doc_c3b00838#page=1						
	Toughened display glass panels	2400 kg	EF-D26-MATERIALUSE-026	1.976	high	
doc_580aafda#page=1						
	A4 copy paper and toner cartridges	780 kg	EF-D26-WASTEDISPOSAL-096	0.004	high	

6. Uncertainty statement

Level	Records
low	0
medium	2
high	18

Main uncertainty sources: document extraction confidence on scanned
factors rather than supplier-specific data, and unit conversions between

1. Request primary activity data and factor references from the three lar
2. Standardize document formats with recurring logistics providers so ex
3. Complete the missing required fields currently parked in the review q
resolved.

7. Reduction scenario advice

Scenario	Reduction t CO2e	Reduction %	
Transport mode shift	0.005	0.00%	Transport mode shift: replacing air and express legs cutting 0.005 t CO2e (0.00% of the baseline). Assu UK Government conversion factor set and cover eq replacement ratio can be tuned between 0% and 10 actual feasibility depends on route schedules, servi this tool cannot verify; cold-chain and time-critical fl averages, so site-specific results will differ; seconda of scope.
Supplier swap	182.136	29.84%	Supplier swap: re-sourcing the activity of the three h Shenzhen CoolTech) to the lowest-value factor avai 182.136 t CO2e. Assumptions: the replacement fact factor set and cover equivalent service levels; activit between 0% and 100% and only the replaced share schedules, service contracts, supplier availability an and time-critical flows may not tolerate slower mode differ; secondary effects such as packaging change
Material substitution	0.805	0.13%	Material substitution: switching high-emission mater the same material family for 1 record(s), abating 0.8 are kept unchanged and listed individually. Assumpt UK Government conversion factor set and cover eq replacement ratio can be tuned between 0% and 10 actual feasibility depends on route schedules, servi this tool cannot verify; cold-chain and time-critical fl averages, so site-specific results will differ; seconda of scope.

8. Appendix I — Emission factors used

ID	Name	Value	Unit	Year	Source
EF-D26-WTTFUELS-015	Fuel upstream (WTT) — Liquid fuels / Aviation spirit — Well-to-tank (t)	813.26357	kgCO2e/t	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-REFRIGERANTOTHER-004	Refrigerant — Kyoto protocol products / HFC-23 — Refrigerant leak (kg)	12400	kgCO2e/kg	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-BUSINESSTRAVELAIR-018	Business travel air — Flights / First class — Flights excl. radiative forcing (RF) (p.km)	0.27701	kgCO2e/p.km	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-E25-FUEL-045	Fuel — Other Oil (>401 deg F) (Stationary Combustion, per MMBtu)	76.46300000000001	kgCO2e/MMBtu	2025	US EPA GHG Emission Factors Hub, January 2025
EF-D26-FUELS-007	Fuel — Gaseous fuels / LPG — Combustion (t)	2939.36095	kgCO2e/t	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-WASTEDISPOSAL-113	Waste — Plastic /				

	Plastics: PET (incl. forming) — Disposal: closed-loop recycling (t)	4.65358	kgCO2e/t	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-WASTEDISPOSAL-010	Waste — Construction / Asphalt — Disposal: landfill (t)	1.27043	kgCO2e/t	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-WASTEDISPOSAL-111	Waste — Plastic / Plastics: LDPE and LLDPE (incl. forming) — Disposal: landfill (t)	9.00687	kgCO2e/t	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-WASTEDISPOSAL-006	Waste — Construction / Average construction — Disposal: combustion (t)	4.65358	kgCO2e/t	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-WTTFUELS-031	Fuel upstream (WTT) — Liquid fuels / Marine fuel oil — Well-to-tank (t)	714.86545	kgCO2e/t	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-WTTFUELS-017	Fuel upstream (WTT) — Liquid fuels / Burning oil — Well-to-tank (t)	660.82605	kgCO2e/t		

				2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-SEAFREIGHT-034	Freight — Cargo ship / 0–4999 dwt — Sea shipping (direct combustion) (t.km)	0.01407	kgCO2e/t.km	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-WTTROADFREIGHT-017	Freight upstream (WTT) — WTT- HGV (refrigerated, all diesel) / Average refrigerated rigids — Diesel (t.km)	0.05168	kgCO2e/t.km	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-WASTEDISPOSAL-080	Waste — Metal / Metal: mixed cans — Disposal: open-loop recycling (t)	4.65358	kgCO2e/t	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-BUSINESSTRAVELAIR-019	Business travel air — Flights / International, to/from non-UK / Average passenger — Flights incl. radiative forcing (RF) (p.km)	0.14253	kgCO2e/p.km	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-MATERIALUSE-070	Materials — Paper / Paper and board: paper — Recycled material, closed-loop (t)	1048.59114	kgCO2e/t	2026	UK Government GHG Conversion

					Factors for Company Reporting 2026 (DESNZ)
EF-D26-DELIVERYVEHICLES-012	Delivery vehicle — HGV (non-refrigerated, all diesel) / Average non-refrigerated HGVs — Diesel van (km)	0.89743	kgCO2e/km	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-WASTEDISPOSAL-134	Waste — Paper / Paper and board: mixed — Disposal: composting (t)	9.00687	kgCO2e/t	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-MATERIALUSE-026	Materials — Other / Glass — Recycled material, closed-loop (t)	823.18954	kgCO2e/t	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-WASTEDISPOSAL-096	Waste — Plastic / Plastics: average plastic film — Disposal: open-loop recycling (t)	4.65358	kgCO2e/t	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)

9. Appendix II — Citation index

Data point	t CO2e	Reference
Total emissions	610.455	calculation-set:task_0c40ed56
Scope breakdown: Scope 1	160.186	aggregation:task_0c40ed56 (Scope 1)
Scope breakdown: Scope 2	208.784	aggregation:task_0c40ed56 (Scope 2)
Scope breakdown: Scope 3	241.485	aggregation:task_0c40ed56 (Scope 3)
Record rec_fc534c9c: Diesel fuel (average biofuel blend) for own fleet, combustion	11.386	record:rec_fc534c9c (doc_cb4cf26e#page=1)
Record rec_1c09b706: Refrigerant HFC-32 top-up for plant HVAC	148.8	record:rec_1c09b706 (doc_293da9de#page=1)
Record rec_009b4f1b: Factory grid electricity, H1 2026	0	record:rec_009b4f1b (doc_4c587592#page=1)
Record rec_e8fb1c36: District heat supply for office campus (2,880 GJ converted)	208.784	record:rec_e8fb1c36 (doc_a1463419#page=1)
Record rec_4cb5b625: Aluminium sheet 5052 for chassis panels	27.924	record:rec_4cb5b625 (doc_5c95444c#page=1)
Record rec_7d426969: PET film, food grade	0.006	record:rec_7d426969 (doc_37858e03#page=1)
Record rec_36b5be57: PP injection pellets	0.003	record:rec_36b5be57 (doc_37858e03#page=1)
Record rec_04d03e5b: LDPE packaging film	0.008	record:rec_04d03e5b (doc_37858e03#page=1)
Record rec_f7ab8df7: 4-layer PCB assemblies (12 lots)	0.003	record:rec_f7ab8df7 (doc_f805bf37#page=1)
Record rec_88a1f5e6: CNC machining centre VM-850 (capital equipment)	5.862	record:rec_88a1f5e6 (doc_6d3c4595#page=1)
Record rec_7f16236c: Well-to-tank upstream emissions of fleet diesel (matches D-01 load)	9.252	record:rec_7f16236c (doc_21c4bca0#page=1)
Record rec_e19008ac: FCL ocean freight of sensor modules, Yantian to Hamburg	4.868	record:rec_e19008ac (doc_429d057d#page=1)
Record rec_636240de: Inbound road freight, Guangzhou to Nanshan (ambient rigid truck)		

	0.046	record:rec_636240de (doc_2e0c5faa#page=1)
Record rec_d4132c91: Landfill disposal of mixed production waste	0.02	record:rec_d4132c91 (doc_b8dcb9c1#page=1)
Record rec_9f8d2ef6: Return economy flight SZX-HAJ for trade fair, 1 passenger	2.509	record:rec_9f8d2ef6 (doc_ced2c2aa#page=1)
Record rec_3738e75f: Heat-treated wooden export pallets (ISPM-15)	0.545	record:rec_3738e75f (doc_1bbf2968#page=1)
Record rec_c4467fcc: Employee commuting rail season tickets, aggregate H1 2026	188.46	record:rec_c4467fcc (doc_42f74b9b#page=1)
Record rec_28d4f911: International express courier, Shenzhen to Austin	0.001	record:rec_28d4f911 (doc_b03c3c53#page=1)
Record rec_279d71a6: Toughened display glass panels	1.976	record:rec_279d71a6 (doc_c3b00838#page=1)
Record rec_c7fef7db: A4 copy paper and toner cartridges	0.004	record:rec_c7fef7db (doc_580aafda#page=1)
S-1 reduction	0.005	scenario:task_0c40ed56_S-1
S-2 reduction	182.136	scenario:task_0c40ed56_S-2
S-3 reduction	0.805	scenario:task_0c40ed56_S-3

10. Disclaimer

This tool output does not constitute an audit opinion or legal advice. Users are advised to perform their own verification before external disclosure.

Excluded record	Description	Reason